

Mastering Prompt Engineering via the IDEA Framework

Workshop Guide & Assessment Rubric

Facilitator Edition

Revision 2 • Professional Development for K–12 and Higher Education

How to Use This Guide

This is a **facilitator** document. It contains everything needed to deliver, assess, and follow up on the workshop, including material that participants should not see in advance. Sections are labeled by audience:

- **Facilitator** — preparation, delivery notes, and scoring procedure.
- **Participant** — content delivered in session; safe to extract into a handout.
- **Administrator** — diagnostic and reporting guidance for instructional leaders.

Before delivering: read §5 (Privacy Compliance Gate) and confirm your district’s approved-tool list and data privacy agreement status. That verification changes what you are permitted to tell participants in §6.

Framework Attribution

The IDEA framework—Include PARTS, Develop CLEAR prompts, Evaluate and REFINE, Apply with accountability—and its component acronyms are the work of **Park and Choo (2024)**, published in *Intervention in School and Clinic*. This guide adapts their framework for professional development delivery and adds the assessment rubric (§5), the transfer-and-measurement structure (§7), and the applied workshop agendas (§§3–4). Extensions original to this guide are marked as such throughout.

Revision 2 changes: framework attribution corrected and made explicit; metric claims qualified against source; linguistic techniques reframed as testable options; rubric rebuilt with a privacy gate and an output-quality dimension; transfer and measurement structure added; reference list added.

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1. Introduction: The Strategic Imperative of AI Literacy

Participant

Generative AI (GenAI) represents a shift in how we interact with technology. We have moved beyond the paradigm in which AI functioned as a static “problem solver” toward a reality in which it can serve as a co-collaborator capable of individualizing the learning experience (Park & Choo, 2024). Prompt engineering is the steering mechanism for that shift, moving our engagement from automation based on association toward purposeful, high-fidelity interaction.

This transition asks educators to treat prompt engineering not as a technical hurdle but as a natural language programming skill—a competency that lets non-technical educators direct complex systems using ordinary prose. The IDEA framework provides the pedagogical bridge, moving educators from unstructured trial and error toward a repeatable methodology that aligns AI capabilities with instructional intent.

A Note on Models

This guide refers to “the AI” as a convenience, but tools differ. Techniques that improve output on one model may do nothing on another, and reasoning-oriented models increasingly perform internal step-by-step work without being asked. Wherever this guide describes a technique, treat the underlying instruction as: *test it on the tool you actually use*.

2. Foundations: The IDEA Framework (Park & Choo, 2024)

Participant

To move beyond casual chatting and toward deliberate design, educators need a structure that enforces intentionality. Park and Choo’s (2024) IDEA framework offers a repeatable methodology for doing so, organized as four sequential moves.

The Four Pillars of IDEA

Pillar	Component	Definition & Action
I — Include PARTS	Persona, Aim, Recipients, Theme, Structure	Define the AI’s role (“Act as a 5th-grade math specialist”), state the goal, describe the audience (Lexile band, IEP considerations), set the topic and tone, and dictate the output format.
D — Develop CLEAR prompts	Concise, Logical, Explicit, Adaptive, Restrictive	Use direct language in a logical order. <i>Restrictive</i> means framing the request within a specific content domain so the model stays inside the boundaries you set.
E — Evaluate & REFINES	Rephrase key words, Experiment with context and examples, Feedback loop, Inquiry questions, Navigate by iterations, Evaluate and verify outputs	Audit the output and revise. Use feedback loops, questioning, and contextual examples (“shots”) to correct errors and close the gap between what you got and what you wanted.

Pillar	Component	Definition & Action
A — Apply with accountability	Ethics, bias, and accuracy	Verify factual claims to catch fabrications. Audit outputs for bias. Disclose AI involvement in instructional materials where required.

Two Different CLEAR Frameworks

Park and Choo’s CLEAR ends in **Restrictive**. A separate and widely cited framework by Lo (2023) uses the same acronym but ends in **Reflective**. These are different constructs, not variations of one: Park and Choo’s R bounds the content domain during drafting, while Lo’s R is a metacognitive evaluation step. If participants arrive having seen Lo’s version—common among those who have done library-led AI literacy training—name the difference rather than letting them assume one is a typo. In IDEA, the reflective function lives in the E pillar.

Why This Matters

The value of a structured approach is that it makes prompting diagnosable. When an output misses, a framework lets you locate *where* it missed—an unstated audience, an unbounded domain, a skipped verification—rather than starting over and hoping for better luck.

Two indicators from the research literature are useful for naming what “better” means, with an important caveat about their maturity:

- **Goal fidelity.** Ali et al. (2023) used goal fidelity as one of three quality indicators—alongside cognitive guidance and social-emotional support—to assess how well a chatbot’s interactions aligned with the goal of a given learning approach. *Extension original to this guide:* we borrow the term to describe how closely an AI output matches the educator’s original instructional intent. This broader usage is ours, not Ali et al.’s.
- **Template stickiness.** The proportion of outputs that conform to a pre-determined format and include the labels specified in the prompt. The term originates with Clavié et al. (2023) and reaches education audiences through Lee and Palmer’s (2025) systematic review.

Caveat: These Are Emerging Indicators, Not Validated Metrics

Lee and Palmer’s (2025) review found that evaluation methods across the prompt engineering literature were largely subjective and that no universal metric of success emerged. Goal fidelity and template stickiness are useful *vocabulary* for structured conversation about output quality. They are not standardized instruments, they have no established benchmarks, and they should not be reported as though they were. Where this guide uses them, treat them as locally defined and locally scored.

3. Short-Format Workshop: 45–60 Minutes

Facilitator

A time-constrained session cannot deliver all seventeen sub-elements of IDEA. This agenda teaches

PARTS thoroughly, **CLEAR** functionally, and **REFINE** as a habit, and defers the full **REFINE** and accountability treatment to the long format or to follow-up. Attempting more produces recall of an acronym without the ability to use it.

60-Minute Agenda

Time	Segment	Content & Facilitator Notes
0–5	Setup & access	Confirm everyone is logged into an approved tool. Budget this time honestly; account friction is the single most common cause of a workshop running short on practice.
5–12	Priming & vocabulary	Define large language models. Introduce tokens (roughly 750 words per 1,000 tokens) and temperature . <i>Say explicitly:</i> temperature is a parameter exposed in API tools and playgrounds but hidden in most consumer chat windows. We cover it so you understand why identical prompts produce different outputs, not because you will be adjusting a slider. Where a tool offers “precise” versus “creative” modes, that is the consumer-facing analog.
12–30	The PARTS of a prompt	Model “setting the scene” on a discipline-specific task (e.g., 5th-grade math word problems at a stated Lexile band). Participants write one PARTS prompt for a task they actually have to do this week.
30–48	Hands-on live remix	Compare zero-shot (no examples) against few-shot (three to five examples). Participants rewrite their own prompt using CLEAR language, run both, and compare. The comparison is the lesson.
48–57	One REFINE cycle	Each participant runs a single documented revision on their output and names what changed and why. This is the minimum viable dose of the E pillar.
57–60	Accountability close	Privacy floor, the human-in-the-loop mandate, and how to submit the Day 30 artifact (see §7).

Linguistic Variables Worth Testing

Participant

The techniques below circulate widely in practitioner communities. Treat them as **hypotheses to test against your own task, not settings that reliably work**. Reported effects tend to be specific to a particular model and a particular kind of problem, and they often fade as models are updated. **None of them substitutes for reading the output and revising it yourself.**

Variable	Reported effect	How to test it
Temperature	Higher values increase randomness and variety; lower values produce more deterministic, conventional output. An API parameter, not a consumer-facing control in most chat tools.	Run the same prompt at two settings if your tool exposes it. If it does not, note where the tool offers precise versus creative modes.

Variable	Reported effect	How to test it
“Take a deep breath”	Reported to improve step-by-step reasoning on math tasks. Originated as a machine-discovered prefix on a single benchmark; transfer to other tasks is unproven.	Run your prompt with and without it. Keep it only if <i>your</i> output improves.
“This is important for my job”	Reported modest gains in response effort and detail. Replication has been mixed.	Same A/B test. Discard if you cannot see a difference.
“Explain your answer”	The most defensible of the four, because its value is not a performance boost—it surfaces the model’s reasoning so <i>you</i> can audit it.	Use routinely on any output you intend to put in front of students.

The Point of This Exercise Is Not the Techniques

It is that you now have a method for evaluating any prompting claim you encounter—including the claims in this workshop. Test, compare, keep what works for your task, discard the rest. That habit is the durable skill, and it is the E pillar in miniature.

4. Long-Format Workshop: Three Hours

Facilitator

The half-day format uses active learning and structured collaborative inquiry to reach the parts of IDEA the short format cannot.

Module 1 — Advanced Engineering (60 minutes)

Focus: chain-of-thought prompting and task decomposition.

Action: Participants design a complex artifact—a multi-day field trip plan, a bilingual oral exam—and instruct the model to work step by step, articulating its reasoning at each stage. Surfacing the reasoning lets educators intervene at the specific point of failure rather than discarding the whole output.

Facilitator Caution

Reasoning-oriented models increasingly perform step-by-step work internally, and explicitly instructing them to “think step by step” can be redundant or occasionally counterproductive. Have participants test both versions on the tool they actually use. The durable skill in this module is *decomposition*—breaking a large task into inspectable stages—not the particular phrasing.

Module 2 — The REFINE Laboratory (60 minutes)

Focus: the iterative loop and structured questioning.

Action: Generate a high-stakes output, then interrogate it. Participants question the model’s rea-

soning, supply counter-examples, and identify misalignments between what they asked for and what they received. Each cycle is documented: what changed, why, and whether it helped. Documentation is what separates iteration from re-rolling.

Module 3 — Custom Instructions & Discipline Integration (60 minutes)

Focus: persistent personas.

Action: Configure custom instructions to establish a durable professional context (“I teach high school chemistry using the 5E model”). Participants audit how a persistent persona changes discipline-specific output, and identify where it helps versus where it over-constrains.

5. The IDEA Assessment Rubric

Facilitator • Administrator

Step 0 — Privacy Compliance Gate (Pass / Stop)

This is not a scored dimension. It is a gate.

If a submitted prompt contains personally identifiable student information, IEP or 504 content, health data, or identifiable grades entered into a tool *without an active district data privacy agreement*, **scoring stops**. The submission receives no rubric score, the participant is redirected to district data-governance training, and the incident is handled under existing privacy protocol.

Rationale: a FERPA violation is a compliance floor, not a developmental stage. A rubric that scores it as “Emerging” implies a teacher can be partially in violation of federal privacy law. They cannot.

Facilitator note: verify your district’s approved-tool list and DPA status before the session. “Never use student data” is the correct default for consumer tools and the wrong instruction if your district has licensed a tool under a signed agreement. Participants will disregard a rule they can see is overbroad.

Scored Dimensions

Component	Emerging (1)	Proficient (2)	Expert (3)
PARTS Alignment	Two or more elements missing. Aim or Recipients left implicit.	All five present. Persona and Aim are unambiguous.	Persona is precisely matched to the task’s disciplinary and pedagogical demands (e.g., “veteran 5th-grade math interventionist using CRA sequencing”) rather than merely elaborate. Recipients specify measurable learner characteristics.
CLEAR Execution	Wordy or ambiguous. No domain restriction; format left to the model.	Concise and logically ordered. Output format stated explicitly.	Restrictive parameters bound content domain, scope, and length. Uses precise professional terminology (Lexile band, 5E phase, DOK level) the model can act on.
REFINE Iteration	First output accepted as-is. No iteration visible.	At least one documented revision cycle correcting a factual, format, or tone problem.	Multiple targeted cycles. Uses inquiry questions to interrogate the model’s reasoning and experiments with examples to close the gap to intent. Revision trail shows <i>why</i> each change was made.

Component	Emerging (1)	Proficient (2)	Expert (3)
Accountability & Verification	No verification attempted. No disclosure.	Checks for obvious factual errors. Discloses AI involvement where required.	Builds verification into the prompt itself (“cite sources I can check”), validates factual and citation claims against trusted references, and audits output for representational bias before use.
Output Quality <i>(new)</i>	Artifact is not classroom-usable without substantial rework, or would be pedagogically unsound if used.	Artifact is usable with light editing. Content is accurate and appropriately pitched.	Artifact is instructionally sound: aligned to a stated standard or objective, appropriately scaffolded, accessible, and demonstrably improved by the teacher’s own revisions.

Why Output Quality Carries Equal Weight

A technically expert prompt that produces a pedagogically poor worksheet is a failure of this workshop’s actual purpose. Output Quality is the only dimension that asks whether any of this helped a student. Do not weight it below the others, and do not treat it as optional.

Scoring Reliability

Facilitator

- **Anchor exemplars are required.** Before any coach or administrator scores independently, calibrate on three pre-scored submissions per dimension—one at each level. Without anchors, two raters will not agree.
- **Calibration threshold.** Two raters should independently score the same five submissions and reconcile differences before scoring live. Target agreement within one point on every dimension.
- **Score the artifact set, not the prompt alone.** All five dimensions require the prompt, the raw output, *and* the human-revised final version. A prompt submitted without its outputs cannot be scored on REFINE or Output Quality.
- **Coach the lowest dimension.** Composite scores hide the actionable information. Feedback should name one dimension and one next step.

Diagnostic Use for Administrators

Administrator

This rubric is a **local instrument for coaching**, not a validated measure. Lee and Palmer (2025) found no universal metric of success in the prompt engineering literature and noted that existing evaluation approaches were largely subjective. Use the rubric to structure conversation and target support; do not report its scores as though they were normed assessment data, and do not use them in personnel evaluation.

With that scope stated, the rubric supports four diagnostic moves:

- **Review prompt libraries.** Look for evidence of inquiry questions and feedback loops in teachers' saved prompt histories. Their presence or absence is the clearest signal of whether IDEA transferred.
- **Low REFINE scores across a team.** Indicates the workshop delivered structure but not iteration. Respond with coaching on comparative prompting rather than a repeat session.
- **Outputs consistently missing intent.** Check whether Restrictive parameters (domain, format, length, scope) are being used at all. This is the most common and most fixable failure.
- **Format conformity as a local check.** If teachers must produce materials on a required district template, you can informally track what share of outputs arrive in the correct format. Define what counts before you count it, and treat the number as a local signal, not a benchmark.

6. Implementation & Ethical Accountability

Participant

Human critical thinking is the final arbiter of AI-generated content. Educators remain the human in the loop and authorize every generated material before it reaches a student.

Responsibility Checklist

- **Data privacy.** Never enter confidential student information, IEP details, or identifiable grades into a consumer AI tool. *If your district has licensed a tool under a signed data privacy agreement, the permitted uses are defined by that agreement*— check the approved-tool list before assuming either permission or prohibition.
- **Accuracy audit.** Cross-reference AI-generated facts, citations, statistics, and worked solutions against trusted sources. Fabricated citations are the most common and most consequential failure mode; a citation that looks plausible is not a citation that exists.
- **Bias mitigation.** Evaluate outputs for stereotyping, uneven representation, and assumptions embedded in training data.
- **Disclosure.** Maintain transparency with students, families, and colleagues about how AI was used in preparing instructional materials.
- **Readiness alignment.** Ensure implementation aligns with the CoSN framework's six core focus areas: executive leadership, operations, data, technology, security, and risk management.

When Not to Reach for AI

A workshop that shows only successes builds brittle confidence. These are cases where the tool is a poor fit and the professional judgment is to decline:

- **Anything requiring verified factual precision you cannot check.** If you lack the subject expertise to catch an error, you cannot use the output responsibly.
- **Individualized decisions about a specific student.** Placement, eligibility, discipline, and diagnosis are human determinations with legal weight.

- **Tasks where the struggle is the learning.** If generating the artifact is what would have built your own understanding of the content, outsourcing it costs more than it saves.
- **Work where provenance matters.** Letters of recommendation, condolence messages, and family communication about a sensitive matter derive their value from being yours.
- **Situations where verification would take longer than doing it yourself.** A short task with high accuracy demands is often faster unassisted.

7. Transfer, Measurement & Follow-Up

Facilitator • Administrator

Single-session professional development reliably produces enthusiasm and unreliably produces changed practice. The structure below extends the workshop across a semester at low facilitator cost and generates evidence an administrator can actually report.

The 90-Day Arc

Phase	Timing	Participant does	Facilitator does	Evidence
Baseline	At registration	Submits one real prompt already in use, plus its output	Scores against rubric; withholds the score	Pre-measure
Session	Day 0	Attends workshop (60 min or half-day)	Delivers	Attendance plus in-session artifact
First application	Days 1–14	Uses IDEA on one authentic task; saves prompt, raw output, and human-revised final version	Available for asynchronous questions	Revision trail
PLC exchange	Day 21	Brings artifact to department or grade-level meeting; a peer scores it	Provides rubric and a 15-minute protocol	Peer scores; collective participation
Coaching touch	Day 30	20-minute individual or small-group check-in	Coaches the lowest-scoring dimension	Targeted follow-up record
Post-measure	Days 60–90	Submits a second prompt, output, and revision	Scores and returns both scores with commentary	Pre/post comparison

Why the Revision Trail Matters Most

Participants submit **three** artifacts, not one: the prompt, the raw AI output, and **the version they actually put in front of students after their own edits.**

The gap between the second and third is the real measure. A substantial gap with thoughtful edits indicates a teacher exercising judgment. A gap of zero means the output was accepted uncritically, which is a coaching flag regardless of how well-constructed the prompt was.

What This Lets You Report

Administrator

- Mean rubric change, pre to post, by dimension
- Percentage of participants moving from Emerging to Proficient on REFINE—the dimension most predictive of continued independent use
- Percentage of submitted outputs showing substantive human revision
- Number of privacy-gate stops, pre versus post (target: zero at post)

Minimum Viable Version

If a full 90-day arc is not feasible, the single highest-value addition is the **Day 30 artifact submission with the revision trail**. One collection point converts the workshop from an event into a measured intervention.

8. Closing

Participant

Prompt engineering asks you to do something you already do well—specify what you want, for whom, at what level, in what form—in a new medium. The framework is not a set of magic words. It is a way of making your existing professional judgment legible to a tool that has no judgment of its own.

That judgment is the part that does not get automated. Models will keep improving at interpreting vague requests, and some of the drafting techniques in this guide will become unnecessary. What will not become unnecessary is knowing whether the output in front of you is good enough for a student to see. Keep testing, keep revising, and keep your name on the final version.

References

- Clavié, B., Ciceu, A., Naylor, F., Soulié, G., & Brightwell, T. (2023). Large language models in the workplace: A case study on prompt engineering for job type classification. *arXiv*. [Verify final published venue before circulating.]
- Lee, D., & Palmer, E. (2025). Prompt engineering in higher education: A systematic review to help inform curricula. *International Journal of Educational Technology in Higher Education*, 22(1), 7. <https://doi.org/10.1186/s41239-025-00503-7>
- Lo, L. S. (2023). The CLEAR path: A framework for enhancing information literacy through prompt engineering. *The Journal of Academic Librarianship*, 49(4).
- Park, J., & Choo, S. (2024). Generative AI prompt engineering for educators: Practical strategies. *Intervention in School and Clinic*. <https://doi.org/10.1177/01626434241298954>

Citations Requiring Completion Before Circulation

The following sources are referenced in the current draft but were not verified during this re-

vision. Retrieve the primary source, confirm the author-date, and complete the entry. Do not circulate a document that mandates a hallucination audit while carrying unverified citations.

- **Ali et al. (2023)** — source of *goal fidelity*. Reached this guide via Lee and Palmer (2025). Retrieve the original.
- **Cain** — cited in the draft as 2024; Lee and Palmer cite Cain (2023). Confirm which.
- **Coy (2023), Chang (2023), Qian (2025)** — confirm full entries.
- **Victa** — appears without year or initial. Identify the source type (peer-reviewed, webinar, or practitioner blog) and label it accordingly in text.
- **CoSN framework** — add the specific publication and year for the six core focus areas.
- **“Take a deep breath” and “This is important for my job”** — if you keep these in §3, cite the originating studies so participants can evaluate the evidence themselves. That is the behavior the section is teaching.